PEER REVIEW HISTORY

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ARTICLE DETAILS

TITLE (PROVISIONAL)	Protocol of a systematic review and network meta-analysis for the prevention and treatment of perinatal depression.
AUTHORS	Smith, Robert; Hung, Sze Chai; Goh, Joyce; Ip, Hoi Lam; Fong, Daniel; Ali, Shehzad; Wilson, Claire; Lok, Kris Yuet-Wan

VERSION 1 – REVIEW

REVIEWER	Kroll-Desrosiers, Aimee Department of Veterans Affairs, Research
REVIEW RETURNED	23-Mar-2021

GENERAL COMMENTS	Note the acronym for surface under cumulative ranking area (SUCRA) is incorrect in the abstract. Many definitions of perinatal depression include onset of depressive symptoms up to a year postpartum. You may want to note this in your Introduction, as 4 weeks is a relative short time postpartum to be addressing, and justify the use of 4 weeks vs. 1 year, especially for the treatment of PND/PPD. A careful read for grammar and clarity is recommended; please also double-check the correct use of punctuation, particularly in lists within sentences. Example: page 12, lines 2-6, there should be a colon [rather than a semi-colon] after 'two different stages: 1.)' and a semi-colon before '2.)'.

REVIEWER	Rollè, Luca University of Turin
REVIEW RETURNED	06-Apr-2021

GENERAL COMMENTS	It was a pleasure for me to read this interesting and complex paper. The manuscript is clear and easy to read. Thanks to the authors for their work. Below some comments and suggestions that I hope can be useful: - please re-write the abstract creating the section discussion/conclusion; - please aff the Prisma Flowchart; - please create a table with the study included; - please create a Conclusion Paragraph; - please specify the language of the papers; - please specify the years considered; - please refers to the PRISMA-P Table in the paper. Best regards.

REVIEWER	Jidong, Dung

	Nottingham Trent University
REVIEW RETURNED	27-Apr-2021

GENERAL COMMENTS	This protocol for systematic review and meta-analysis is timely and intended to address two specific research objectives: (a) to identify the optimal approach for preventing perinatal depression in women and (b) to identify the optimal approach for the treatment of perinatal depression in women. Overall, this study seems to be sophistically designed and intended to be executed with rigorous procedures. Introduction: (i) It would be beneficial to include some introductory cultural variation in the aetiology, diagnosis and intervention for perinatal depression. (ii) There are missed citations in places (e.g. Line 6 “Clinical guidelines recommend screening and treatment for perinatal depression” and line 19-21 “Many treatments evaluated were identified as having a positive effectiveness on depression symptoms compared to usual care or placebo, but the relative comparability of these treatment has not previously been investigated.”). (iii) A more focused approach would be beneficial (e.g. a systematic review and meta syntheses that explore a particular culture/context/country or region). A universal or general approach could be too vague to be articulated in a single protocol/study. Method: a. Are/Is there other supporting information regarding this protocol (e.g. PROPERO registration number)? b. Can authors provide some rationale and/or justification of why search teams were taken from previously published meta-analyses? Also, kindly provide some justification for the chosen databases. General comments: Please provide concrete example(s) that would help delineate and illustrate more effective ways of bridging and linking anticipated findings to practical relevance and implications for clinical practice and research. Thorough proofreading copyediting with the appropriate academic tone would be beneficial. There were typos and grammatical errors in place that potentially distorts the authors’ intended meanings.

VERSION 1 – AUTHOR RESPONSE

Reviewer 1 Comments:

R1-1: Many definitions of perinatal depression include onset of depressive symptoms up to a year postpartum. You may want to note this in your Introduction, as 4 weeks is a relative short time postpartum to be addressing, and justify the use of 4 weeks vs. 1 year, especially for the treatment of PND/PPD.

R1-1 Response:
Whilst conducting initial pilot testing of identifying and screening of articles we found that there are a number of studies that included postpartum depression beyond 4 weeks that were relevant to our systematic review. We agree with the reviewers comments that perinatal depression could include onset depression and therefore took the decision to revise our inclusion criteria to include those up to one year postpartum. To reflect this we have made revisions to
the abstract, introduction, methods and Table 1.

Page 2 Line 22: “Depression experienced during pregnancy and after birth, also known as perinatal depression”

Page 3 Line 17: “Identifying those with depression early; either during the antenatal period (during pregnancy) or in the postnatal period (up to 1 year postpartum) provides a critical opportunity”

Page 6 Line 14: “Briefly, studies on participants who are perinatal women between 20 weeks gestation to 1 year after birth will be included. We selected 1 year after birth to reflect the time period in which there is risk of postpartum depression occurring between day 1 to 1 year postpartum[32].”

R1-2: A careful read for grammar and clarity is recommended; please also double-check the correct use of punctuation, particularly in lists within sentences. Example: page 12, lines 2-6, there should be a colon [rather than a semi-colon] after 'two different stages: 1.)' and a semi-colon before '2.)'.

R1-2 Response:
We have conducted checking for grammar, corrected use of punctuation and have made further changes to the writing of the manuscript to improve readability and clarity. These changes throughout the manuscript have been highlighted in yellow within the highlighted changes version.

Reviewer 2 Comments:

R2-1: Please re-write the abstract creating the section discussion/conclusion;

R2-1 Response:
Thank you for your suggestion, we have structured our abstract following the BMJ Open author guidelines for protocol manuscripts which asks abstracts to the structured with the following sections. “Introduction; Methods and analysis; Ethics and dissemination. Registration details should be included as a final section, if appropriate”

For this protocol manuscript we shall maintain the recommended structure within our abstract.

R2-2: Please aff the Prisma Flowchart;

R2-2 Response:
As this is a protocol for a planned systematic review study we do not currently have any results in terms of study selection to include in this protocol manuscript. When we complete the planned study we shall ensure to report on the study selection process using the PRISMA flowchart.

R2-3: Please create a table with the study included;

R2-3 Response:
In line with our response to our previous comment we do not yet have any results from the systematic review in order to report a table of included studies' characteristics. When we complete the planned study we shall report on the characteristics of included studies in full.

R2-4: Please specify the language of the papers; Please specify the years considered;

R2-4 Response:
Thank you for your comment, we included all studies written in English, regardless of year of publication. To address this in the manuscript we have included an additional sentence in the eligibility criteria subheading of the methods section.

Page 7 Line 4: "Included articles were limited to those written in English, there was no limitation of publication year for included articles."

R2-5: please refers to the PRISMA-P Table in the paper.

R2-5 Response:
We have included a reference to the PRISMA-P in the methods sections to illustrate the protocol development following these guidelines.

Page 6 Line 8: "The protocol of this study has been developed following the Preferred Reporting Items for Systematic Reviews and Meta-Analyses Protocols (PRISMA-P) guidelines (appendix 1)"

Reviewer 3 Comments:

R3-1: It would be beneficial to include some introductory cultural variation in the aetiology, diagnosis and intervention for perinatal depression.

R3-1 Response:
To address this we have introduced some context for perinatal depression among different cultures within the introduction section of the manuscript.

Page 3 line 1: "Previous systematic reviews have shown that the prevalence of perinatal depression is generally higher in low-to-middle-income countries than high-income countries in both the antenatal and postnatal stages [3 4]. With perinatal mental disorders, including depression, being more prevalent in mothers who are the most socioeconomically disadvantaged [4]. Cultural factors also have been indicated to be sources of inequality for perinatal mental illness, these include cultural gender-bias, gender-biased violence; both physically and mentally [4].

R3-2: There are missed citations in places (e.g. Line 6 "Clinical guidelines recommend screening and treatment for perinatal depression" and line 19-21 "Many treatments evaluated were identified as

having a positive effectiveness on depression symptoms compared to usual care or placebo, but the relative comparability of these treatment has not previously been investigated.").

R3-2 Response:
The missing citation in the first case is referred to in the following sentence, to improve clarity, we have combined both sentences into one and maintained the citation to the NICE clinical guidelines at the end of this sentence. To address the second missing citation we have added the relevant citations of Stephens et al. (2016) & Howard & Khalifeh (2020).

Page 4 Line 16: "Clinical guidelines recommend screening and treatment for perinatal depression, these guidelines do not provide recommendations on which treatments are most effective[26]"

Page 4 Line 22: "but the relative comparability of these treatment has not previously been investigated[24 27]."

R3-3: A more focused approach would be beneficial (e.g. a systematic review and meta syntheses that explore a particular culture/context/country or region). A universal or general approach could be too vague to be articulated in a single protocol/study.

R3-3 Response:
We agree that a general approach of network meta-analysis may introduce bias towards intervention types that have more published studies in certain countries or cultures compared to others, potentially affecting generalisability of our findings for particular cultures or regions. However, we would prefer not to be selective in the studies and data we include in our review based on where they were conducted, as this can reduce the amount of evidence would make use of, reducing precision for interventions with relatively fewer published studies, it could also further reinforce bias towards countries and cultures with more published articles addressing perinatal depression (if we were to select a focus on the regions with the most published articles). To address this issue we shall plan to conduct a meta-regression and sub-group analysis, separating studies by region and conducting a network meta-analysis in each region and report on the findings separately, identifying commonality or differences among regions allowing us to evaluate generalisability.
We have already considered other factors for which treatments for perinatal depression may vary and have planned to address this with meta-regression and sub-group analysis, the geographical region studies are conducted will also be added as a factor to be explored during analysis.
To reflect this we have added the following:

Page 12 Line 7: "Factors for exploring in the meta-regressions shall include year of publication, the geographical region the study was conducted, study sample age, whether the study participants were at antenatal or early postnatal stage of motherhood, study sample's baseline depression severity (if possible), risk of bias in all 5 domains and overall risk of bias."

R3-4: Are/Is there other supporting information regarding this protocol (e.g. PROPERO registration number)?

R3-4 Response:
In response to the editor's comment above, we have included the correct PROSPERO registration number of the study. Following reviewer 2's comment, we have referred to the PRISMA-P table in the text as supporting information.

Page 2 Line 5 “Prospero ID number CRD42020200081”

Page 5 Line 18: “The protocol of this study has been developed following the Preferred Reporting Items for Systematic Reviews and Meta-Analyses Protocols (PRISMA-P) guidelines (appendix 1)”

R3-5: Can authors provide some rationale and/or justification of why search teams were taken from previously published meta-analyses? Also, kindly provide some justification for the chosen databases.

R3-5 Response:
We selected general health sciences databases MEDLINE, EMBASE, WoS as well as Cochrane database based on the recommendation for covering health related research articles within the Cochrane handbook for systematic reviews. We selected BNI, CINAHL Plus and PsycINFO as specialist databases that may include relevant studies within perinatal depression. Identified relevant search terms from previous systematic review to maximise sensitivity of our search terms identifying all definitions of perinatal depression and minimise risk to missing relevant studies. To address this we added the following:

Page 7 line 10: “We selected MEDLINE, EMBASE and WoS databases based on recommendations for Cochrane Handbook covering major health sciences topics. We also selected specialist subject databases BNI, CINAHL Plus and PsycINFO based on their relevance to the study objectives.”

Page 7 line 22: “We identified all search terms, related to the two sets of PICOS, from previously published meta-analyses on the prevention or treatment of perinatal depression[15-25] to maximise sensitivity of our search terms identifying all definitions of perinatal depression and minimise risk to missing relevant studies. An example full list of search terms for the MEDLINE database can be found in appendix 2.”

R3-6: Please provide concrete example(s) that would help delineate and illustrate more effective ways of bridging and linking anticipated findings to practical relevance and implications for clinical practice and research.

R3-6 Response:
To provide a more concrete example of how anticipated findings of this study can lead to implications for clinical practice, we have given an illustration in the rationale section of the manuscript. We illustrate how a network meta-analysis inform health care by comparing to providers on trade-off in clinical effectiveness between interventions that are low in resources to resource intensive interventions.

Page 5 line 15: “An example of this could be making relative comparisons on the effectiveness of interventions that require fewer resources for health providers to implement that is scalable against more resource intensive interventions that require trained specialist or equipment to implement and the level of trade off in clinical effectiveness between those interventions. Another advantage of Bayesian network meta-analysis is statistical certainty can be estimated, this allows for identifying potentially promising interventions with low levels of statistical certainty that may require further investigation to establish effectiveness.”

R3-7: Thorough proofreading copyediting with the appropriate academic tone would be beneficial. There were typos and grammatical errors in place that potentially distorts the authors’ intended meanings.

R3-7 Response:

We have conducted checking for grammar, corrected use of punctuation and have made further changes to the writing of the manuscript to improve readability and clarity. These changes throughout the manuscript have been highlighted in yellow within the highlighted changes version.